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Interrelations between sea hub ports and inland hinterlands: Perspectives of multimodal freight transport organization and low carbon emissions

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ABSTRACT

In 2020, a "dual carbon" goal was proposed by China's government. As a low-pollution and low-consumption transportation organization mode, multimodal freight transport has been highly valued. Strengthening the organization and management of multimodal freight transport between seaports and inland areas is conducive to reducing carbon emission levels and logistics costs in the transportation industry. In the multimodal freight transport network, a reasonable hinterland for the main hub port can effectively avoid homogeneous competition, realize the optimal allocation of transport resources, and promote the integrated and coordinated development of the region. In constructing a multimodal freight transport network and arranging a reasonable organization mode, it is beneficial to divide the hinterland of main hub ports and promote energy conservation and emission reduction in the transportation industry. From the perspective of hinterland division and coordination of transportation organization methods, we construct a 0–1 planning model to minimize total cost, transportation time, and carbon emissions. By combining the nondominated ranking method and ideal point method, the algorithm sifts and reorganizes each transportation mode and path in different cities, thereby further overcoming the subjectivity of the weighting method and obtaining the nondominated Pareto optimal frontier solution set. Finally, the evaluation function is used to determine the optimal scheme. Taking the main hub port in the Yangtze River Delta region as an example to verify the validity of the model and algorithm, this research can provide decision support for departments when they formulate policies related to optimizing the regional transportation structure, can promote regional integration and coordinated development of coastal and inland areas, and has practical significance for the "dual carbon" goal.

1. Introduction

In 2015, the Chinese government released the "Enhanced Action on Climate Change - China's National Contribution", which announced the government's pledge to achieve peak CO_2 emissions by approximately 2030, and in 2020, the government announced its efforts to achieve carbon neutrality by 2060, thus forming China's "dual carbon" target. The transportation sector accounts for approximately 10% of China's carbon emissions, and the Chinese government and local authorities have formulated action plans to reduce carbon emissions from the transportation sector. Multimodal freight transport is an effective way to link sea hub ports with inland hinterlands and is an intensive and efficient transportation organization mode that is highly valued by the Chinese government. Ports are the core nodes of multimodal freight transport, and strengthening the organization and management of port collection and distribution can effectively improve transport efficiency and reduce transport costs and carbon emissions. Studies on port-centered multimodal freight transport focus mainly on the transport organization between ports and hinterlands, the rational hinterland division of main hub ports and optimizing the transport network. These factors contribute to the integrated and synergistic development between ports and hinterlands.

After the 14th Five-Year Plan, China will enter a new era of the inland economy. Cities located in a better radiating location in the interior, especially those cities with water ports and major rail access, will play a major role in future economic growth. As globalization and regional integration continue to grow, the strategic position of major global city clusters has improved, the interaction between port clusters

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and urban clusters has become more frequent, and both types of clusters have continued to gradually integrate into an organic whole (Celliers et al., 2020; Papaheochari and Coccossis, 2019). In recent years, China has promulgated a series of plans and policies to promote coordinated regional development. The regional coordinated development system and mechanism have been improved. These efforts will provide strong support for promoting high-quality development.

In the context of regional integration, there is homogeneous competition among ports within the region (de Miranda Grilli et al., 2019; Wei et al., 2021). For example, the Yangtze River Delta, an important international hub for China's economy, plays an important role in the external attraction capacity of the Yangtze River Economic Belt. Due to their geographical proximity, similar cargo sources, and duplication in infrastructure construction, Shanghai Port and Ningbo-Zhoushan Port (the main hub ports in the Yangtze River Delta region) have problems, such as unreasonable multimodal freight transport organization and hinterland overlap. Additionally, the proportion of railroad transportation in the two major ports is significantly low: the proportion is only 0.3% in Shanghai Port and only approximately 0.8% in Ningbo-Zhoushan Port. Unoptimized transportation methods lead to a waste of resources; therefore, the overall functions of the integrated transportation system cannot be fully utilized.

Multimodal freight transport, as a mode of transport organization in the integrated transport system, features low cost, low pollution, and high efficiency, which can effectively optimize the transport structure and break through the bottleneck of economic development. Additionally, multimodal freight transport can promote the formation of a unified market with interconnections between coastal areas and inland ones and realize the "double-cycle" pattern at home and abroad. The inhomogeneity of the transport structure requires us to fully exploit the technical and economic advantages of different modes of transportation, to clarify the advantages of different modes of transportation in each region and their functions and positioning in the development of multimodal freight transport and to build suitable transportation infrastructure. Otherwise, there might be redundant construction and a considerable waste of transportation resources. The rational division of the transport network structure between the main hub port and the hinterland can realize the optimal allocation of transport resources in the region and avoid homogeneous competition in the region.

The main aim of this paper is to realize the interconnection between coastal areas and inland areas and enhance the complementarity of the main hub ports. This research makes four major contributions:

(1) A multiobjective 0–1 planning model is established with three goals, and an algorithm combining the nondominant sorting method and ideal point method is designed to achieve these goals.
(2) Taking the main hub port in the Yangtze River Delta region of China as an example, the hinterland division of the main hub port is realized, and the optimal solution containing the transport path and mode is determined. The calculation also proves the effectiveness of the model algorithm.
(3) This paper provides policy-related suggestions for departments: to strengthen the construction of inland port stations and to adjust the transport structure of the freight market through price leverage. Thus, this paper has good theoretical and practical application value.
(4) Our research on the interrelations between sea hub ports and inland hinterlands is instructive for multimodal freight transport organization and management and policy formulation.

The remainder of this paper is organized as follows. We review the related literature in Section 2. Section 3 shows the cooperative optimization model of the multimodal freight transport network, and Section 4 describes the solving algorithm. Case analysis is presented in Section 5. We conclude the paper and suggest directions for future work in Section 6.

2. Literature review

The research involves mainly the coordination mechanism of transport organization and the optimization of the multimodal freight transport network.

Orderly operation helps the port improve its competitiveness. Macroeconomics, technological innovation and policy all greatly affect the positioning and development of ports (Xu et al., 2021). Wang et al. (2021) emphasized the power of innovation in marine science and technology. Z. Zhang et al. (2018) studied the EOQ-based sea-rail logistics network in the context of the Internet of Things. With respect to collaboration among transportation organizations, Filippova et al. (2021) established multimodal freight transport and logistics centers to manage cargo flow distribution. Verma et al. (2012) indicated that drayage accounts for a significant portion of transport risk and that this risk can be reduced by arranging direct trains. In algorithm design, heuristic algorithms (such as exact solution algorithms and forbidden search algorithms) are preferred (Crevier B. et al., 2012). Simulation experiments using software such as Cplex and MATLAB are also used (Bhattacharya et al., 2014. Xie et al., 2012).

In terms of multimodal freight transport network optimization, the optimization objectives include mainly minimal factors, such as transportation time, transportation costs, and carbon emissions (Ding, 2020. Islam et al., 2016. Q. Z. Wang et al., 2020). Considering the mixed time window constraints and the changes in the transportation network, the mixed integer programming model is established from a dynamic perspective (Van Riessen et al., 2015). Concerning algorithm design, there are shortest-circuit algorithms, ant colony algorithms, genetic local search algorithms, and other heuristic algorithms. (Cho et al., 2012. FU et al., 2017). In addition, M. Zhang et al. (2015) take container transportation in the Dutch hinterland as an example to study the freight optimization model. Ricci et al. (2016) use the case of Munich-Riem in Germany to evaluate the impact of innovative measures on intermodal freight terminals.

In China, the scale distribution of major ports is relatively stable, with cargo throughput tending to be scattered, while container throughput tends to be concentrated. (Xu et al., 2021). The hinterland division of the major hub ports in the coastal region helps to allocate resources more rationally. Due to the overlap of city influences, the impact of spatial boundaries in traditional hinterland research has gradually weakened. Hinterland studies should focus on the intensity of urban impacts from a geographic scale perspective (Cheng and LeGates, 2018). Concerning method selection, there are actual survey verification methods, mathematical simulation methods, spatial software analysis and network association measurement methods (Zhao et al., 2014. Tao et al., 2016).

The existing studies have three problems. First, studies on multimodal freight transport synergistic optimization from a regional perspective lack systematic results. Second, too few studies combine hinterland division with multimodal freight transport networks for synergistic optimization. Third, multiobjective optimization solution algorithms for multimodal freight transport networks can be further improved (See Table 1).

In contrast with existing studies in related areas, our model has the following characteristics. First, we combine hinterland division with transportation networks and modes to build a multimodal freight transport synergy optimization model consisting of multiple transportation nodes and different transportation modes. With the objectives of minimizing transportation costs, time, and carbon emissions, we determine how the cargo flow of each hinterland should be transported to the corresponding main hub port. In this way, the container multimodal freight transport network can be optimized. Second, concerning algorithm design, we use the nondominated sorting method to obtain the Pareto optimal frontier solution and then use the evaluation function

Table 1
Summary of relevant studies.

Question category	Author & Year of Publication	Methodology	Contribution
Organization Management	Xu et al., 2021	Applies the rank-size rule.	The grade scale distribution of major coastal port systems in China is described to provide reference for government departments at all levels to formulate port planning.
Organization Management	Wang et al. (2021)	Constructs panel fixed effect model and random effect model.	Suggested to strengthen the innovation power of marine science and technology, improve innovation funding support, deepen international cooperation, etc.
Organization Management	Wan, C. et al. (2021)	The centrality metric of the network is incorporated into the TOPSIS framework as a way to evaluate port importance.	Based on the entropy weight TOPSIS method, the top ten important ports of MSR shipping network were evaluated to verify the validity of the model.
Organization Management	Ding, L. et al. (2020)	Proposes an improved cross efficiency model incorporating undesirable outputs.	Presented the trends and characteristics of regional development in the coastal areas of China.
Collaboration of transportation organizations	F. Nadezhda et al., 2021	Establishes multimodal freight transport and a logistics center.	Studied how to find optimal transport solutions for distributing cargo flows in a given area.
Collaboration of transportation organizations	Z. Zhang et al. (2018)	Proposes a method of logistics network reliability analysis based on EOQ in the Internet of Things environment.	Reduced the cost of sea-rail intermodal transportation and improved the stability of intermodal transportation.
Collaboration of transportation organizations	Verma et al. (2012)	Proposes a biobjective optimization framework and develops a tabu-search based solution methodology.	Indicated that drayage accounts for a significant portion of transport risk and that risk can be reduced by arranging direct trains.
Collaboration of transportation organizations	Crevier B. et al., 2012	Proposes a new bilevel mathematical formulation.	An exact solution method based on the mixed integer formulation is proposed, and randomly generated examples are computed.
Collaboration of transportation organizations	Bhattacharya et al. (2014)	Proposes a model for strategic transport planning involving a network-wide intermodal transport system.	Successfully applied the model to an existing Fast Moving Consumer Goods (FMCG) distribution network in India.
Collaboration of transportation organizations	Xie et al. (2012)	Proposes a multimodal HAZMAT model that simultaneously optimizes the locations of transfer yards and transportation routes.	Applied the model to two case studies of different network sizes to demonstrate its applicability and provided suggestions for future research.
Multimodal freight transport network optimization	Ding (2020)	A model considering the constraints of mixed time windows is established, and the ant colony algorithm is used to optimize the network.	Established an electronic platform suitable for information interconnection and the interoperability of multimodal freight transport stations.
Multimodal freight transport network optimization	Islam et al. (2016)	Adopts literature survey and group discussion methods and uses systematic methods to conduct corresponding research.	Facilitated the transition from road transport to rail transport in the European transport market.
Multimodal freight transport network optimization	Q. Z. Wang et al. (2020)	The efficiency evaluation framework is established by combining order relationship analysis and an entropy weight fuzzy analytical hierarchy method.	Validated the proposed approach using the China-Europe railway network from Wuhan, China, to Berlin, Germany.
Multimodal freight transport network optimization	Van Riessen et al. (2015)	A Linear Container Allocation model with Time-restrictions is proposed.	Proposed that network operators can select priority areas to prevent high-impact interference and improve planning in the case of high interference correlation.
Multimodal freight transport network optimization	Cho et al. (2012)	Presents a Weighted Constrained Shortest Path Problem (WCSPP) model and uses a dynamic programming algorithm to optimize the network.	Applied multiple Pareto optimal solutions to a mathematical model and MADM model and compared the two evaluation methods.
Multimodal freight transport network optimization	FU et al. (2017)	Establishes a dynamic analysis model with double objectives and uses an ant colony algorithm to optimize the network.	Obtained the optimal route recommendation plan of Wuhan-Europe international container multimodal freight transport based on customer needs.
Multimodal freight transport network optimization	M. Zhang et al. (2015)	Presents a model that simultaneously incorporates multimodal infrastructure, hub-based service network structures, and the various design objectives.	Policy packages combining multiple types of policies show better network performance as compared with the optimal performance resulting from a single policy type.
Multimodal freight transport network optimization	Ricci et al. (2016)	Uses a methodological approach for evaluation; this approach includes other analysis	Assessed the possible impact of different innovative operational measures on the intermodal freight station.

(continued on next page)

Table 1 (*continued*)

Question category	Author & Year of Publication	Methodology	Contribution
		methods and simulation models.	
Hinterland Research	Cheng and LeGates, 2018	Analyzes three key indicators of the complex network model: network density, network degree centrality, and network hinterland.	Shanghai is considered to be the command and control center of the Yangtze River Delta, with increasing organization and management of its hinterland.
Hinterland Research	Zhao, M. et al., 2014	Based on the partition core algorithm, an improved method for network analysis of a single company in the entire calculation process is proposed.	Avoided information loss due to the merging of two network data in the original algorithm.
Hinterland Research	Tao et al., 2016	Combines the network correlation method and relative correlation method.	Reflected the spatial linkage of the cross-regional network in terms of mobility while considering the regional nature of the network hinterland.

of the ideal point method to obtain the optimal solution in the set. The algorithm overcomes the subjectivity of the weighting method and avoids the problem of difficult selection due to the complex Pareto solutions. Finally, we take the Yangtze River Delta region of China as an example to verify the validity of the model. The model not only realizes the hinterland division of the main hub ports in the Yangtze River Delta region but also avoids homogeneous competition and realizes the reasonable allocation of transportation resources.

3. Cooperative optimization model of multimodal freight transport network

3.1. Problem description

From the perspective of the main hub port hinterland division and coordination of transportation organization methods, we build a multimodal freight transport network synergy optimization model composed of starting points, transit points, destinations, and different transportation methods with the goal of minimizing transportation costs, time, and carbon emissions. In the collaborative optimization model, the transportation modes between nodes include railway, waterway, and road transportation. According to the results, the attraction range of the main hub port is determined, and the transport modes between the main hub port and each hinterland node are also determined.

3.2. Assumptions

(1) It is assumed that there is a linear relationship between transportation cost and transportation distance.
(2) It is assumed that cargo transit occurs at most once to reduce transit costs, time, and carbon emissions.

3.3. Symbol definition

See (Table 2).

3.4. Objective function

(1) Minimum transportation cost: min C

The entire transportation cost includes the cost of transportation from the starting point to the transit point and from the transit point to the destination and the transportation cost for reloading. Due to the long distance between the hinterland and the port, the transportation costs incurred during transferring and loading account for a small proportion of the entire transportation cost and can be negligible. The transportation costs during transfer are included in c_{ij}^{w} and c_{jk}^{w}. Transport-related costs can be expressed as:

$$\min C = \sum_{i}\sum_{j}\sum_{w} c_{ij}^{w} q_{ij} x_{ij}^{w} + \sum_{j}\sum_{k}\sum_{g} c_{jk}^{w} q_{jk} x_{jk}^{w} \tag{1}$$

(2) Minimum carbon emissions: min E

The full transportation carbon emissions include the carbon emissions from the transportation origin to the transit node and from the transit node to the transportation destination and the carbon emissions from the transit process. The carbon emissions during transfer are included in e_{ij}^{w} and e_{jk}^{w}. The total carbon emissions can be expressed as:

$$\min E = \sum_{w}\sum_{i}\sum_{j} e_{ij}^{w} q_{ij} x_{ij}^{w} + \sum_{g}\sum_{j}\sum_{k} e_{jk}^{w} q_{jk} x_{jk}^{w} \tag{2}$$

(3) Minimum transit time: min T

The total transportation time includes the transportation time from the transportation origin to the transfer node and from the transfer point to the destination and the transfer time for reloading at the transfer point. The time consumption during the transfer process is considered in t_{ij}^{w} and t_{jk}^{w}. The total time can be expressed as:

$$\min T = \sum_{w}\sum_{i}\sum_{j} t_{ij}^{w} q_{ij} x_{ij}^{w} + \sum_{g}\sum_{j}\sum_{k} t_{jk}^{w} q_{jk} x_{jk}^{w} \tag{3}$$

3.5. Restrictions

$$\sum_{w}\sum_{i} x_{ij}^{w} \le 1, \quad j = 1, 2... \tag{4}$$

$$\sum_{w}\sum_{j} x_{jk}^{w} \le 1, \quad k = 1, 2... \tag{5}$$

$$Q_i \le \sum_{j}\sum_{w} Q_{ij}^{w} x_{ij}^{w} \tag{6}$$

$$\sum_{i} q_{ij} = \sum_{k} q_{jk}, \forall j \tag{7}$$

$$\sum_{i} Q_{ij}^{w} x_{ij}^{w} + Q_j \le \sum_{k}\sum_{g} Q_{jk}^{w} x_{jk}^{w} \tag{8}$$

Constraint (4) means that for each transfer node, only one mode can be selected for transport to one destination. Constraint (5) means that for each transportation origin, only one mode can be selected for transport to one transfer point. Constraint (7) ensures the balance of the incoming and outgoing cargo flow. Constraints (6) and (8) ensure that the cargo throughput capacity on each route can meet the cargo flow at the original delivery place.

4. Solving algorithm

Generally, a multiobjective planning model is transformed into a

single-objective problem by artificially setting weight coefficients and then solved by a heuristic algorithm. The usual transformation methods are the linear weighted sum method and the ideal point method (Y. Wang et al., 2019). In the linear weighted sum method, corresponding weight coefficients are assigned to each objective according to its importance. Since the importance of different objectives is difficult to quantify, the distribution of weight coefficients is not supported by sufficient data, and its objectivity is insufficient.

For double objective problems, the Pareto optimal front is usually a curve in the two-dimensional coordinate system. For multiobjective problems, the Pareto optimal front is usually represented as a hypersurface in space. In this paper, we study the multiobjective optimization problem, and the hypersurface determined by the set of nondominated solutions is believed to be the Pareto optimal front. In practical processes, we first calculate the values of three objectives of all schemes, perform nondominated sorting on the values, and eliminate the schemes whose values are dominated by the other values. We obtain the first-level nondominated solution set in this way. Finally, each solution in the set is analyzed using the evaluation function to obtain an optimal solution. The specific flow of the algorithm is shown in Fig. 1.

4.1. Evaluation index

In the nondominated solution set, many Pareto frontier solutions can be found. Each solution in the set is better than the others in one specific objective. Not one solution in the set is superior in all objectives. The optimal solution selection can be found by setting the evaluation function, such as the crowding degree operator in the NSGA II algorithm. The ideal point method is a multi-indicator evaluation method that measures the strengths and weaknesses with the help of how close the evaluation object is to the idealized target. Compared with the crowding degree operator, the ideal point method evaluation function can overcome the uneven distribution of individuals and low efficiency of the Pareto algorithm. In this paper, we use the evaluation function of the ideal point method for evaluation. The evaluation function is expressed as follows:

$$y=\left(\left(\sum_i\sum_j\sum_w c_{ij}^w x_{ij}^w+\sum_j\sum_k\sum_g c_{jk}^g x_{jk}^g-f_1^*\right)^2+\left(\sum_w\sum_i\sum_j e_{ij}^w x_{ij}^w+\sum_g\sum_j\sum_k e_{jk}^g x_{jk}^g-f_2^*\right)^2+\left(\sum_w\sum_i\sum_j t_{ij}^w x_{ij}^w+\sum_g\sum_i\sum_j t_{jk}^g x_{jk}^g-f_3^*\right)^2\right)^{1/2} \quad (9)$$

In this function, the evaluation indicators f_1^*,f_2^* and f_3^* represent the optimal values of the single objectives of minimum cost, minimum carbon emissions, and shortest time, respectively. Each part of this evaluation index represents the distance between the actual value and optimal value of a single objective in each scheme. The optimal solution is the one whose actual values of the three objectives are closest to the ideal value. Therefore, the solution with the smallest evaluation index is the optimal solution.

4.2. Solution steps

Step 1: Input relevant data, and write the objective function and constraint function program.

Step 2: Calculate the optimal value under each single goal; the goals are denoted as f_1^*, f_2^*, and f_3^*.

Step 3: Use the nondominated sorting algorithm to stratify the population size to obtain the first-level nondominated layer of the population.

(1) Set $i = 1$.

(2) For all $j = 1, 2, \ldots, n$, and j is not equal to i, according to the definition of the nondominated solution, compare the dominance and

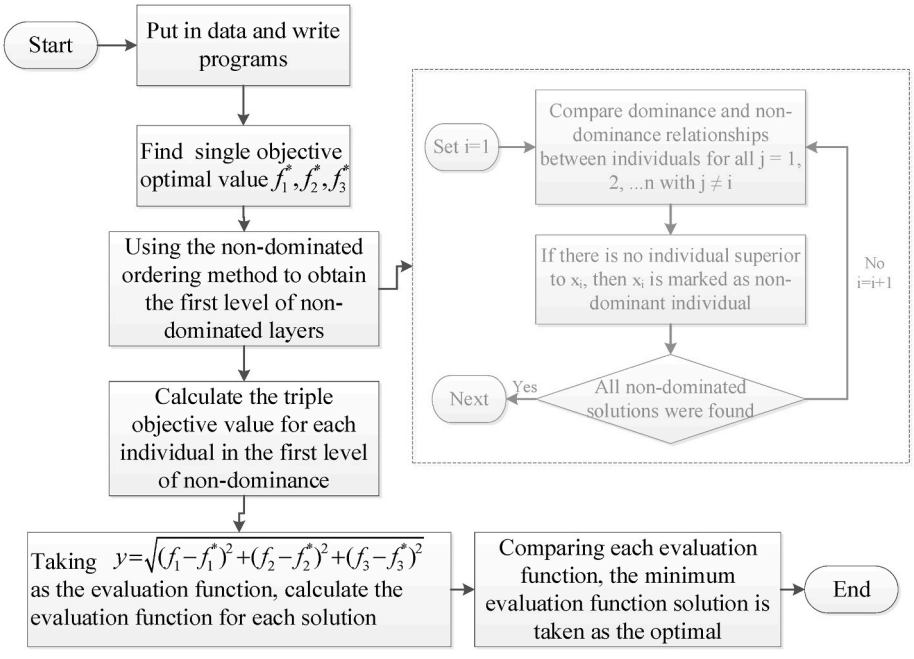


Fig. 1. Algorithm flow chart.

nondominated relationship between individual$x_i$and individualx_j.

(3) If no individual x_j is better than x_i, then x_i is marked as a nondominated individual.

(4) Let$i = i + 1$; go to step 3.2 until all nondominated individuals are found.

Step 4: Calculate the three goals of each nondominated individual in the first-level nondominated layer.

Step 5: Taking $y = \sqrt{(f_1 - f_1^*)^2 + (f_2 - f_2^*)^2 + (f_3 - f_3^*)^2}$ as the evaluation function, substitute the specific f_1^*, f_2^*, and f_3^* values obtained in the previous step into the evaluation function, and calculate the evaluation function corresponding to each nondominated solution.

Step 6: Compare the value of the evaluation function corresponding to each solution in the first-level nondominated solution set, and take the solution with the smallest evaluation index as the optimal solution for the multiobjective problem.

Table 2
Definition of symbols.

Sets	Definition
I	Set of transportation destination nodes, $i \in I$.
J	Set of transport transit nodes, $j \in J$.
K	Set of transportation starting nodes,$k \in K$.
M	Target collection, $m \in M$; in this collection, 1 represents freight, 2 represents carbon emissions, and 3 represents transportation time.
W	Set of transportation methods, $w \in W$; in this set, 1 denotes railway transportation, 2 denotes water transportation, and 3 denotes road transportation.
Q	Total cargo weight (ton).
Parameters	
c_{ij}^w	The sum of unit transportation costs and transit costs between nodes i and $j$through the $w-$ th transportation method (yuan/TEU).
c_{jk}^w	The sum of unit transportation costs and transit costs between nodes$j$and k through the $w-$ th transportation method (yuan/TEU).
t_{ij}^w	The transportation time from node i to node j through the $w-$ th transportation mode.
t_{jk}^w	The transportation time from node j to node $k$through the $w-$ th transportation mode.
e_{ij}^w	The unit carbon emissions of the $w-$ th transportation mode between node i and node j (kg/TEU).
e_{jk}^w	The unit carbon emissions of the $w-$ th transportation mode between node j and node k (kg/TEU).
d_{ij}^w	The transportation distance of the $w-$ th transportation mode between node i and node j (km).
d_{jk}^w	The transportation distance of the $w-$ th transportation mode between node j and node k (km).
q_{ij}	The freight volume from node i to node j.
q_{jk}	The freight volume from node j to node k.
Q_{ij}^w	The maximum transportation capacity of the $w-$ th transportation mode between node i and node j.
Q_{jk}^w	The maximum transportation capacity of the $w-$ th transportation mode between node j and node k.
f_m	The value of target $m(m \in M)$ corresponding to the current solution.
f_m^*	The optimal value of single target $m(m \in M)$.
Variables	
$x_{ij}^w = \begin{cases} 1 \\ 0 \end{cases}$	The value of transportation mode w when the mode is selected between transportation node i and transportation node j is 1; otherwise, the value is 0.
$x_{jk}^w = \begin{cases} 1 \\ 0 \end{cases}$	The value of transportation mode w when the mode is selected between transportation node j and transportation node k is 1; otherwise, the value is 0.

Table 3
Case node data.

I	Shanghai Port, Ningbo-Zhoushan Port
J	Virtual Shanghai transfer station, virtual Ningbo-Zhoushan transfer station, Wuhan, Zhengzhou, Changsha
K	Urumqi, Chengdu, Wuhan, Xi'an, Chongqing, Changsha, Nanchang, Zhengzhou, Lanzhou

5. Case study

5.1. Case description

Currently, the Yangtze River Delta port dislocation development situation has been formed. In terms of the specific port division of labor, the Yangtze River Delta region has formed a "One Body Two Wings" port group with Shanghai Port and Ningbo-Zhoushan Port as the main ports and Jiangsu, Zhejiang, and Anhui along the river coast as the two wings. In this paper, we take Shanghai Port and Ningbo-Zhoushan Port as the main hub ports. Urumqi, Chengdu, Wuhan, Xi'an, Chongqing, Changsha, Nanchang, Zhengzhou, and Lanzhou, which are the hinterlands of the two main hub ports, are selected as the starting points based on the reference of cargo volume and cargo turnover. Wuhan, Zhengzhou, and Changsha, which have strong location advantages, superior geographical location, and large cargo transit volume, are selected as transit points to constitute a complete cargo transportation network. As a result, a practical transportation network has been formed. In this network, the cargo can be transported directly or in transit. To indicate direct transport, we set a virtual transit point in the model; if the virtual transit point is selected, the actual transport is indicated as direct transport. The detailed node data are shown in Table 3.

5.2. Data collection

The railway freight data in this article come from the China Railway Network and China Transportation Yearbook (http://www.stats.gov.cn/tjsj/ndsj/). The waterway freight and highway freight data are obtained by consulting related freight forwarders, and the rail freight rates are obtained from the website of China Railway 95306 (http://www.95306.cn/). Time data are obtained by distance and speed measurement from the official website of AutoNavi Maps (https://www.amap.com/). The carbon emission data are measured by distance and unit carbon emission amount (http://www.nra.gov.cn/). All basic data are detailed in the appendix.

5.3. Solution results and analysis

In this paper, MATLAB is used for solving the model. The optimal value of each objective is first calculated separately and saved as required data in the evaluation function. Then, through the nondominated ranking method, the first level of nondominated layers that are not dominated by other solutions and have no dominating relationship with each other is obtained as the Pareto frontier solution set. As this paper considers the tri-objective optimization model and plots the position of each element in the frontier solution set in three-dimensional space, the Pareto frontier hypersurface can be obtained as shown in Fig. 2. Through calculation, we obtain the three objective values corresponding to each individual in the first-level nondominated layer and the evaluation function corresponding to each solution. The four index values of the 33 specific schemes in the final nondominated solution set are shown in Table 4:

The results show that among inland cities, more cities choose Zhengzhou as a transit station. This indicates that Zhengzhou, as a comprehensive transportation hub in the inland area, plays an important role in the transportation network between the inland areas and the Yangtze River Delta region. Zhengzhou's railway transportation

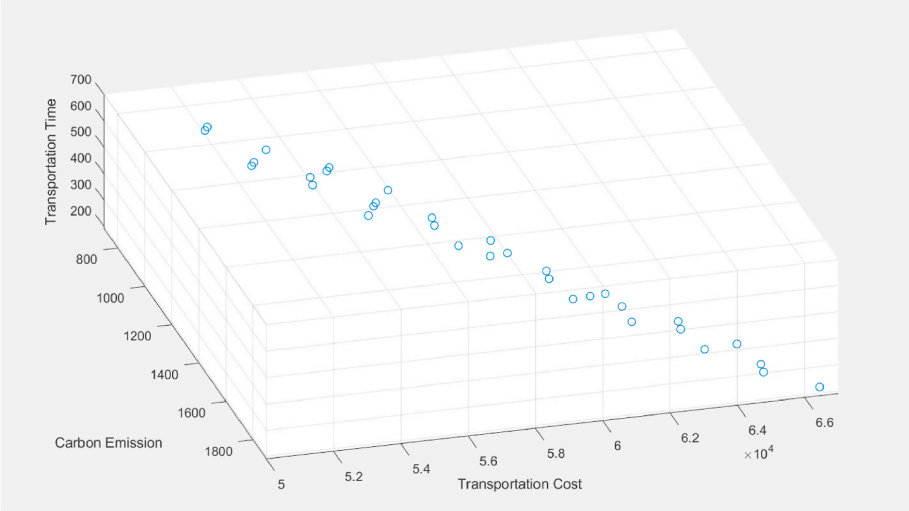


Fig. 2. Multitarget scatter plot.

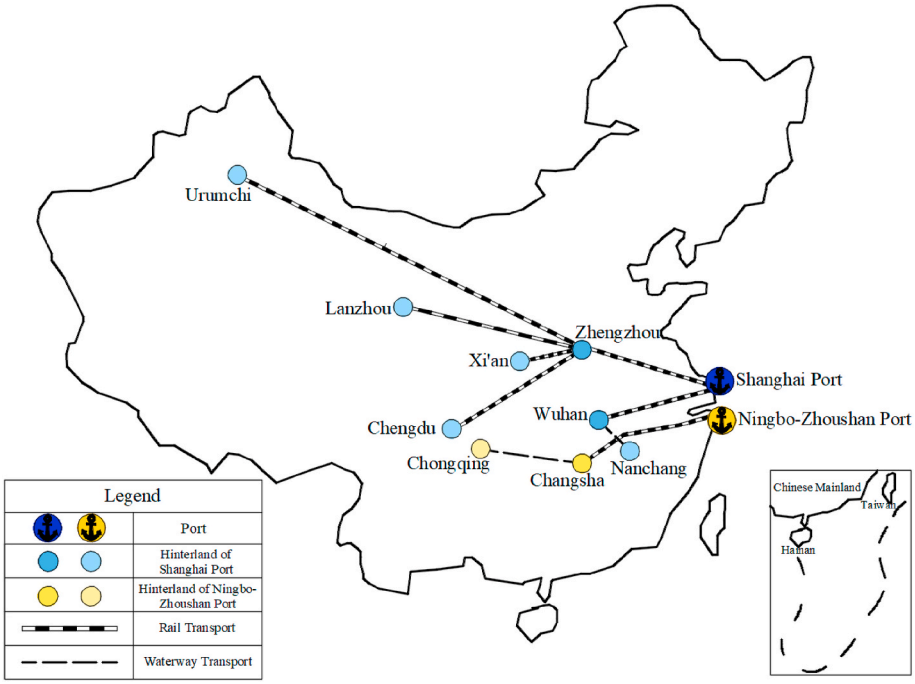


Fig. 3. Optimal transportation network.

capacity provides a strong foundation for cargo transit. In long-distance transportation, railway transportation has become the preferred choice because it has low carbon emissions, it outpaces water transportation, and its cost is lower than the cost of road transportation. In addition, we found that cargoes from Urumqi are transported by rail to Zhengzhou for transit and finally arrive in Shanghai. Changsha, the hinterland of Ningbo-Zhoushan Port, has cargoes transported by rail. In contrast, the cargo transportation mode of Chongqing (which is the hinterland of Ningbo-Zhoushan Port) cannot be determined, and the scheme can be chosen according to the actual pursuit of customers. Finally, with the help of the evaluation function in the ideal point method, we integrated the effects of the three objectives and obtained scheme 17 with the smallest evaluation function. This scheme is noted as the optimal solution and is shown in Table 5.

The solutions obtained using the nondominated sorting combined with the ideal point evaluation function are compared with the solutions obtained by NSGA II (see Fig. 3). The solution sets during the first and second iterations of the two methods are also intercepted and displayed in Figs. 4 and 5 for comparison. We found that the triple objective value corresponding to the solution obtained by combining the ideal point evaluation function is relatively smaller; this result is more consistent with the pursuit of the minimum objective, that is, closer to the Pareto optimum. The result shows that the Pareto efficiency of the solution obtained by iteration exceeds the solution obtained by the traditional NSGA II, thereby indicating that this method is effective.

6. Conclusion

Reasonable multimodal freight transport organization and management can strengthen the interrelations between sea hub ports and inland hinterlands and effectively reduce carbon emissions. Additionally, because ports are the core of multimodal freight transport, the network

Table 4
Transportation plan target value.

Number	Transportation cost (yuan/TEU)	Carbon emission (kg/TEU)	Time (h)	Evaluation function	Number	Transportation cost (yuan/TEU)	Carbon emission (kg/TEU)	Time (h)	Evaluation function
1	57842.2	759.6283	217.9837	12233.54	18	53561.39	1013.555	666.3671	7971.479
2	57675.9	784.0219	224.9837	12067.29	19	59104.55	860.8738	216.719	13496.26
3	60001.3	1068.872	209.361	14395.93	20	61263.65	1170.118	208.0962	15660.34
4	59030.3	898.4462	215.3598	13422.35	21	60292.65	999.6917	214.0951	14685.93
5	61189.4	1207.69	206.737	15587.15	22	62451.75	1308.936	205.4723	16852.01
6	53871.83	846.2478	492.4851	8269.413	23	55134.18	947.4932	491.2203	9532.345
7	53705.53	870.6414	499.4851	8103.811	24	60522.3	1065.923	210.7738	14916.76
8	56030.93	1155.492	483.8623	10434.11	25	62681.4	1375.167	202.151	17083.79
9	55059.93	985.0657	489.8611	9458.926	26	63869.5	1513.985	199.5271	18276.37
10	59311.2	1005.691	212.747	13704.73	27	61991.3	1311.986	205.5371	16391.91
11	61470.3	1314.935	204.1242	15871.32	28	64150.4	1621.23	196.9143	18561.7
12	62658.4	1453.753	201.5003	17063.82	29	65338.5	1760.048	194.2904	19755.14
13	56343.66	788.1177	394.4897	10737.09	30	63943.75	1476.412	200.8863	18349.05
14	56177.36	812.5113	401.4897	10571.07	31	65131.85	1615.23	198.2624	19541.88
15	57531.76	926.9356	391.8657	11926.07	32	65412.75	1722.475	195.6496	19827.43
16	52373.29	874.7371	668.991	6783.084	33	66600.85	1861.293	193.0256	21021.03
17	52206.99	899.1307	675.991	6618.237					

Table 5
Optimal transportation scheme.

Optimal solution						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Waterway	Waterway	Railway
J	Zhengzhou	Zhengzhou	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port

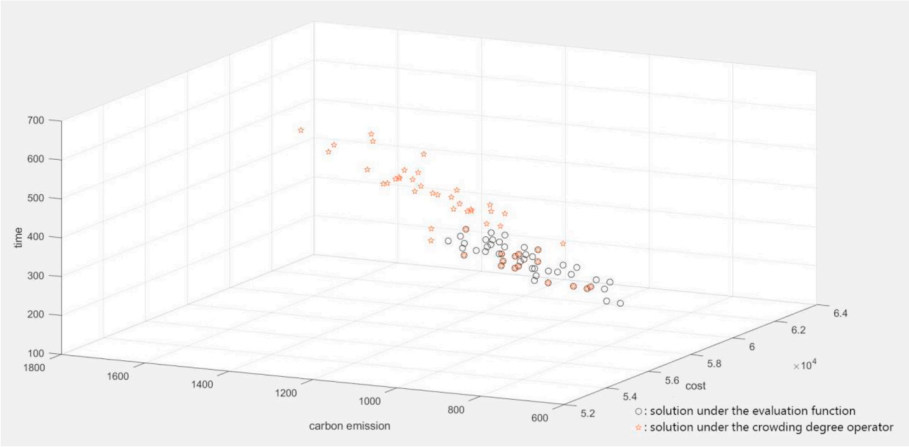


Fig. 4. Solution sets during the first iteration.

optimization of port collection and distribution is an effective way for port management departments to adjust the transport structure. In the context of regional integration, managers can make relevant guiding policies, which can clarify the functions of major hub ports and fully exploit their respective location and economic advantages. Carrying out a reasonable division of labor in transportation modes can effectively avoid homogeneous competition in the region and realize a reasonable distribution of transportation resources while providing customers with reasonable transportation modes and path choices.

This paper constructs a multimodal freight transport system synergy optimization model from the perspective of hinterland division and transport organization mode synergy for the main hub ports in the region. As a result, we optimize the hinterland division and multimodal freight transport organization for cities outside the region. The 33 solutions show that cargoes in Changsha should to be transported by railroad to Ningbo-Zhoushan Port, while cargoes in Wuhan and Zhengzhou should be transported by railroad to Shanghai Port. More cities choose Zhengzhou as a transit station due to its better railroad infrastructure. Finally, we found that long-distance transportation is mostly performed by railroads.

Our algorithm, which is combined with the nondominated ranking method and the ideal point method, has strong practical value for solving multiobjective network optimization problems. The algorithm overcomes the subjectivity of the weighting method and avoids the problem of difficult selection due to the overly complex Pareto solution. In addition, the research results also show that the division of the main hub port's hinterland is related to the location conditions of the hinterland region and the infrastructure level of the transportation mode. Zhengzhou plays an important role in the transportation hub of the inland region due to the city's location advantage and developed railroad transportation system, thus indicating the importance of multimodal freight transport infrastructure and node construction.

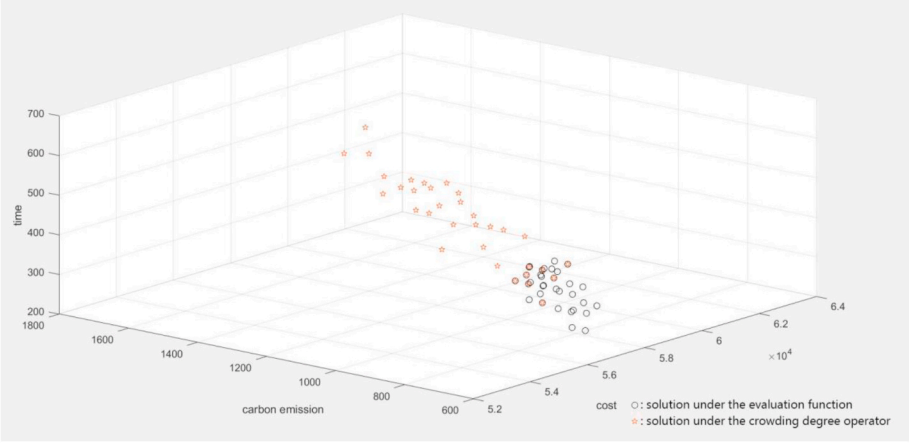


Fig. 5. Solution sets during the second iteration.

Therefore, government departments should coordinate port and inland hub facilities and equipment and strengthen the construction of inland port stations. For example, introducing rail lines to ports and logistics parks can promote seamless intermodal transport systems. In the face of excessive competition in the highway freight market, by enhancing the governance of overloading and overlimiting transportation on highways, road-to-railroad transportation can be promoted, and the advantages of railroad transportation over medium and long distances can be embodied. Additionally, the government can adjust the transport structure in the freight market through price leverage. Because the current railroad freight prices are high and road transport prices are inverted, the development of railroad tariff preferential policies, water transport price subsidy policies, and the corresponding tax policies can help to achieve road-to-railroad and road-to-water transportation. Finally, a multimodal freight transport network with a clearer division of labor, closer cooperation, and smoother connection can promote the formation of a unified market with interconnection between coastal areas and inland ones and realize the "double-cycle" pattern at home and abroad.

From the perspective of enterprises, with the rapid development of China's high-speed rail, the existing line railroad freight capacity has been greatly released, especially in economically developed regions. The railroad transport department develops diversified products for regional integration mainly for road-to-railroad transportation. These products include direct express trains and direct cold chain trains. Shipping enterprises can take advantage of the collection and distribution of inland waters to develop direct transportation routes from the river to sea and achieve road-to-water transportation.

To date, the fluctuation of freight rates over time and the impact of the uncertainty of transportation demand on the optimized structure have not been considered. Additionally, the algorithm can still be further improved. Combining hyperheuristic algorithms, studying the main hub port hinterland division, and multimodal freight transport network optimization from the dynamic perspective are the direction of future research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table 1
Transportation cost data (yuan/TEU)

	Urumqi			Chengdu		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	15740.80	–	–	9267.40	–	–
Ningbo-Zhoushan Port	16089.20	–	–	9561.90	–	–
Wuhan	14688.58	–	–	6985.98	–	9145.08
Zhengzhou	19179.55	–	–	9833.50	–	12700.00
Changsha	19940.80	–	–	11210.60	–	13031.10
	Xi'an			Chongqing		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	6196.70	–	–	8569.10	1600.00	–
Ningbo-Zhoushan Port	6545.10	–	–	8863.70	–	–
Wuhan	5150.68	–	6504.68	6287.88	2068.76	7579.18
Zhengzhou	6656.00	–	7844.10	9590.40	–	12361.90
Changsha	10402.90	–	11666.10	9574.10	5603.73	11043.10
	Nanchang			Lanzhou		

(*continued on next page*)

Table 1 (*continued*)

	Urumqi			Chengdu		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	3649.10	865.00	5007.10	9076.00	–	13987.40
Ningbo-Zhoushan Port	3359.70	–	4676.00	9424.30	–	14961.80
Wuhan	2863.68	1365.14	4126.03	8023.78	–	10836.28
Zhengzhou	8009.20	–	10299.00	9534.80	–	12214.90
Changsha	6742.40	5376.59	7660.00	13281.90	–	16341.40

Table 2
Carbon emission data (kg/TEU)

	Urumqi			Chengdu		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	236.79	–	–	125.39	–	–
Ningbo-Zhoushan Port	243.24	–	–	130.39	–	–
Wuhan	333.92	–	–	202.84	–	512.08
Zhengzhou	236.79	–	–	146.36	–	456.14
Changsha	297.08	–	–	144.97	–	459.74
	Xi'an			Chongqing		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	87.64	–	–	113.20	280.69	–
Ningbo-Zhoushan Port	94.09	–	–	123.59	–	–
Wuhan	187.50	–	387.19	195.28	275.10	438.01
Zhengzhou	87.64	–	226.46	139.45	–	440.15
Changsha	151.41	–	395.18	119.64	206.26	365.71
	Nanchang			Lanzhou		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	46.87	93.73	236.84	126.90	–	661.60
Ningbo-Zhoushan Port	45.19	–	221.17	133.35	–	707.69
Wuhan	147.54	176.03	248.79	226.76	–	592.07
Zhengzhou	109.71	–	342.58	126.90	–	433.20
Changsha	90.43	160.20	181.05	187.19	–	616.32

Table 3
Transportation time data (h)

	Urumqi			Chengdu		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	67.95	–	–	35.98	–	–
Ningbo-Zhoushan Port	69.80	–	–	37.42	–	–
Wuhan	334.70	–	–	297.08	–	288.46
Zhengzhou	67.95	–	–	42.00	–	29.95
Changsha	85.25	–	–	41.60	–	32.14
	Xi'an			Chongqing		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	25.15	–	–	32.48	454.24	–
Ningbo-Zhoushan Port	27.00	–	–	35.47	–	–
Wuhan	292.68	–	284.42	294.91	573.61	286.06
Zhengzhou	25.15	–	22.53	40.02	–	29.43
Changsha	43.45	–	30.05	34.33	308.83	29.10
	Nanchang			Lanzhou		
Transportation method	Railway	Waterway	Road transport	Railway	Waterway	Road transport
Shanghai Port	13.45	235.54	9.30	36.42	–	21.83
Ningbo-Zhoushan Port	12.97	–	8.67	38.27	–	23.32
Wuhan	281.21	457.72	279.95	303.95	–	291.04
Zhengzhou	31.48	–	26.28	36.42	–	29.21
Changsha	25.95	254.95	23.13	53.72	–	37.20

Table 4
Data of transit-main hub port

		Wuhan			Zhengzhou		
Transportation method		Railway	Waterway	Road transport	Railway	Waterway	Road transport
Transportation cost (yuan/TEU)	Shanghai Port	4038.20	1024.38	5769.40	4281.80	–	–
	Ningbo-Zhoushan Port	4170.80	–	6296.50	5131.10	–	–
Carbon emission (kg/TEU)	Shanghai Port	47.10	127.97	272.89	57.96	–	–
	Ningbo-Zhoushan Port	57.09	–	297.82	71.03	–	–
Transportation time (h)	Shanghai Port	13.52	275.60	10.75	16.63	–	–
	Ningbo-Zhoushan Port	16.38	–	11.37	20.38	–	–
		Changsha					
Transportation method		Railway	Waterway	Road transport			
Transportation cost (yuan/TEU)	Shanghai Port	4998.10	1124.34	7564.20			
	Ningbo-Zhoushan Port	4708.80	–	6969.90			
Carbon emission (kg/TEU)	Shanghai Port	68.13	158.39	357.79			
	Ningbo-Zhoushan Port	66.10	–	329.68			
Transportation time (h)	Shanghai Port	19.55	311.18	13.32			
	Ningbo-Zhoushan Port	18.97	–	12.05			

Table 5
33 specific schemes.

Scheme 1						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 2						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
J	Zhengzhou	Zhengzhou	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 3						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Railway	Railway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 4						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Highway	Railway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 5						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Highway	Railway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 6						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Waterway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 7						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Waterway	Railway	Railway
J	Zhengzhou	Zhengzhou	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 8						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Railway	Waterway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 9						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Highway	Waterway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou

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Table 5 (*continued*)

Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 10						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Highway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 11						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Railway	Highway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 12						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Highway	Highway	Railway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 13						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Railway	Waterway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 14						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Railway	Waterway	Railway
J	Zhengzhou	Zhengzhou	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 15						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Highway	Railway	Waterway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme Scheme 16						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Waterway	Waterway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 17						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Waterway	Waterway	Railway
J	Zhengzhou	Zhengzhou	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 18						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Highway	Waterway	Waterway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 19						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Railway	Highway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 20						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Railway	Railway	Highway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 21						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Highway	Railway	Highway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 22						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Highway	Railway	Highway	Railway

(*continued on next page*)

Table 5 (*continued*)

J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 23						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Waterway	Highway	Railway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 24						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 25						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Railway	Railway	Railway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 26						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Highway	Railway	Railway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 27						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Railway	Railway	Highway	Railway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 28						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Railway	Highway	Railway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 29						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Highway	Highway	Railway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 30						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Railway	Railway	Highway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 31						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Highway	Railway	Highway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 32						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Railway	Highway	Highway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port
Scheme 33						
I	Urumqi	Chengdu	Xi'an	Chongqing	Nanchang	Lanzhou
Transportation method	Railway	Highway	Highway	Highway	Highway	Highway
J	Zhengzhou	Wuhan	Zhengzhou	Changsha	Wuhan	Zhengzhou
Transportation method	Railway	Railway	Railway	Railway	Railway	Railway
K	Shanghai Port	Shanghai Port	Shanghai Port	Ningbo-Zhoushan Port	Shanghai Port	Shanghai Port

References

Bhattacharya, A., Kumar, S.A., Tiwari, M., Talluri, S., 2014. An intermodal freight transport system for optimal supply chain logistics. Transport. Res. C Emerg. Technol. 38, 73–84. https://doi.org/10.1016/j.trc.2013.10.012.

Celliers, L., Rosendo, S., Costa, M.M., Ojwang, L., Carmona, M., Obura, D., 2020. A capital approach for assessing local coastal governance. Ocean Coast Manag. 183, 104996.

Cheng, Y., LeGates, R., 2018. China's hybrid global city region pathway: evidence from the Yangtze River Delta. Cities 77, 81–91. https://doi.org/10.1016/j.cities.2018.01.015.

Cho, J.H., Kim, H.S., Choi, H.R., 2012. An intermodal transport network planning algorithm using dynamic programming—a case study: from Busan to Rotterdam in intermodal freight routing. Appl. Intell. 36 (3), 529–541. https://doi.org/10.1007/s10489-010-0223-6.

Crevier, B., Cordeau, J.F., Savard, G., 2012. Integrated operations planning and revenue management for rail freight transportation. Transp. Res. Part B Methodol. 46 (1), 100–119. https://doi.org/10.1016/j.trb.2011.09.002.

Ding, L., 2020. Multimodal transport information sharing platform with mixed time window constraints based on big data. J. Cloud Comput. 9 (1), 11. https://doi.org/10.1186/s13677-020-0153-8.

Ding, L., Yang, Y., Wang, L., Adrian, C.C., 2020. Cross Efficiency Assessment of China's marine economy under environmental governance. Ocean Coast Manag. 193, 105245. https://doi.org/10.1016/j.ocecoaman.2020.105245.

Filippova, N., Bogumil, V., Vlasov, V., 2021. Improving the reliability and safety of cargo delivery for the Arctic through the creation of transport and logistics centers for multimodal transportation management. IOP Conf. Ser. Earth Environ. Sci. 678 (1), 012019 https://doi.org/10.1088/1755-1315/678/1/012019.

Fu, X.-p., He, Y.-s., Zou, M., Fan, D.-f., 2017. A study on the route selection of international container transport in the multimodal scenario. Railway Transp. Econ. 39 (12), 12–17. https://doi.org/10.16668/j.cnki.issn.1003-1421.2017.12.03.

Grilli, N.D.M., Xavier, L.Y., Jacobi, P.R., Turra, A., 2019. Integrated science for coastal management: discussion on a local empirical basis. Ocean Coast Manag. 167, 219–228.

Islam, D.M.Z., Ricci, S., Nelldal, B.-L., 2016. How to make modal shift from road to rail possible in the European transport market, as aspired to in the EU Transport White Paper 2011. Eur. Transp. Res. Rev. 8 (3), 18. https://doi.org/10.1007/s12544-016-0204-x.

Papaheochari, T., Coccossis, H., 2019. Development of a waterfront regeneration tool to support local decision making in the context of integrated coastal zone management. Ocean Coast. Manag. 169, 284–295. https://doi.org/10.1016/j.ocecoaman.2018.12.013.

Ricci, S., Capodilupo, L., Mueller, B., Karl, J., Schneberger, J., 2016. Assessment methods for innovative operational measures and technologies for intermodal freight terminals. Transport. Res. Procedia 14, 2840–2849. https://doi.org/10.1016/j.trpro.2016.05.351.

Tao, L., Rui, Z., 2016. Urban hinterworld in Yangtze River Delta: empirical comparison of two network-based methods. Acta Geograph. Sin. 71 (2), 236–250. https://doi.org/10.1007/s12544-016-0204-x.

Van Riessen, B., Negenborn, R.R., Lodewijks, G., Dekker, R., 2015. Impact and relevance of transit disturbances on planning in intermodal container networks using disturbance cost analysis. Marit. Econ. Logist. 17 (4), 440–463. https://doi.org/10.1057/mel.2014.27.

Verma, M., Verter, V., Zufferey, N., 2012. A bi-objective model for planning and managing rail-truck intermodal transportation of hazardous materials. Transport. Res. E Logist. Transport. Rev. 48 (1), 132–149. https://doi.org/10.1016/j.tre.2011.06.001.

Wan, C., Zhao, Y., Zhang, D., Yip, T.L., 2021. Identifying important ports in maritime container shipping networks along the Maritime Silk Road. Ocean Coast Manag. 211, 105738. https://doi.org/10.1016/j.ocecoaman.2021.105738.

Wang, Y., Olgun, C.G., Wang, L., Meng, B., 2019. Risk assessment of water inrush in Karst Tunnels based on the ideal point method. Pol. J. Environ. Stud. 28 (2), 901–911. https://doi.org/10.1016/j.tust.2013.05.001.

Wang, Q.-Z., Chen, J.-M., Tseng, M.-L., Luan, H.-M., Ali, M.H., 2020. Modelling green multimodal transport route performance with witness simulation software. J. Clean. Prod. 248, 119–245. https://doi.org/10.1016/j.jclepro.2019.119245.

Wang, L., Su, M., Kong, H., Ma, Y., 2021. The impact of marine technological innovation on the upgrade of China's marine industrial structure. Ocean Coast Manag. 211, 105792. https://doi.org/10.1016/j.ocecoaman.2021.105792.

Wei, B., Li, Y., Suo, A., Zhang, Z., Xu, Y., Chen, Y., 2021. Spatial suitability evaluation of coastal zone, and zoning optimisation in ningbo, China. Ocean Coast Manag. 204, 105507.

Xie, Y., Lu, W., Wang, W., Quadrifoglio, L., 2012. A multimodal location and routing model for hazardous materials transportation. J. Hazard Mater. 227, 135–141. https://doi.org/10.1016/j.jhazmat.2012.05.028.

Xu, Q., Huang, T., Chen, J., Wan, Z., Qin, Q., Song, L., 2021. Port rank-size rule evolution: case study of Chinese coastal ports. Ocean Coast Manag. 211, 105803. https://doi.org/10.1016/j.ocecoaman.2021.105803.

Zhang, M., Janic, M., Tavasszy, L., 2015. A freight transport optimization model for integrated network, service, and policy design. Transport. Res. E Logist. Transport. Rev. 77, 61–76. https://doi.org/10.1016/j.tre.2015.02.013.

Zhang, Z., Zhao, J., Cui, J., Nie, H., 2018. Reliability analysis of logistics network of rail-sea intermodal transportation in Internet of Things environment. J. Coast Res. 83, 823–827. https://doi.org/10.2112/si83-135.1.

Zhao, M., Wu, K., Liu, X., Derudder, B.E., 2014. A novel method to approximate intercity networks and its empirical validation. Acta Geograph. Sin. 69 (2), 169–183. https://doi.org/10.11821/dlxb201402003.